Data and Computer Communications

Medium Access Control (MAC) techniques

MAC, MAC protocol

- Technique to access the medium for transmission
- Protocols that enable multiple users to share a transmission

Broadcast N/W and MAC

- Broadcast : Transmit to everyone reachable.
- In any broadcast network, the key issue is how to determine who gets to use the channel when there is **competition** for it.
- In the literature, broadcast channels are sometimes referred to as **multiaccess channels** or **random access channels**.
- When only a single channel is available, determining who should go next is much harder. Many protocols are proposed for solving this problem.

- The protocols used to determine who goes next on a multiaccess channel belong to a sublayer of the data link layer called the **MAC** (**Medium Access Control**) sublayer.

Channel Allocation Problem

- **Static Channel Allocation**
  - Channel characteristic and properties are available at static time.
  - Techniques like FDM, TDM etc are used.
- **Dynamic Channel Allocation**
  - Characteristics in terms of parameters like **Station Model, Collision model** etc are reqd. to be decided dynamically.
  - Collision control.
    - If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a **collision**.
    - All stations can detect collisions. A collided frame must be transmitted again later. There are no errors other than those generated by collisions.

Collision Assumptions

Assumptions for collision control.

- **Continuous Time.**
  - Frame transmission can begin at any instant. There is no master clock dividing time into discrete intervals.
- **Slotted Time.**
  - Time is divided into discrete intervals (slots).
  - Frame transmissions always begin at the start of a slot.
  - A slot may contain 0, 1, or more frames, corresponding to an idle slot, a successful transmission, or a collision, respectively.
- **Carrier Sense.**
  - Stations can tell if the channel is in use before trying to use it.
  - If the channel is sensed as busy, no station will attempt to use it until it goes idle.
- **No Carrier Sense.**
  - Stations cannot sense the channel before trying to use it. They just go ahead and transmit.
  - Only later can they determine whether the transmission was successful.

Objectives

- MAC layer is to define how a user access a channel when he needs one.
  - Random access: ALOHA and CSMA
  - Ordered access: Token bus and Token Ring
  - Deterministic access: FDMA, TDMA, and CDMA
  - Combinations: TDMA-over-FDMA, TDD-CDMA, and TDMA/CSMA

Multiple Access Protocols

- Many MAC protocols are there.
- Many algorithms/techniques are devised for allocating a multiple access channel based on technique to access the medium.
  - ALOHA
  - CSMA
  - CSMA/CD
  - CSMA/CA
  - Token ring
  - Token bus etc
- Randomized access protocols
  - A node transmits **at random** at **full** channel data rate R.
  - If two or more nodes "**collide**", they retransmit at **random times**
  - The **random access MAC** protocol specifies how to detect collisions and how to recover from them (via delayed retransmissions, for example)
  - Example
    - Aloha
    - CSMA variants
    - Classic Ethernet (CSMA/CD)

Multiple Access Protocols

- Standard LAN Protocols:
  - Based on multiple access techniques LAN specific MAC protocols are devised.
    - IEEE 802.X
    - FDDI (Fiber Data Distributed Interface)
    - Satellite networks
    - LAN switching
    - VLAN
    - WLAN
    - PAN
    - WiMax etc

Ground Wave/Radio

- Waves propagation can be classified depending upon the frequencies as
  - Ground waves propagation, Skywave propagation, Free space propagation
- Ground Wave Propagation
  - Also known as **surface waves,** propagates over the earth's surface in low and medium frequencies. Mainly used for transmission between the surface of the earth and the ionosphere. Made up of several constituent waves.
  - It is known as a ground wave because it is the sum of the waves that are reflected by the earth's surface or any obstacles. The waves follow the earth's curvature, enabling them to cover beyond the horizon. Beyond the horizon, the waves get blocked by the earth's curvature, and the signals are produced by the diffracted surface wave
  - Frequency of ground waves depending on the type of ground:

Type of ground	**f = 1 MHz**	**f = 10 MHz**	**f = 100 MHz**
Dry ground like desert	10^{-4}	10^{-4}	10^{-4}
Very moist ground like fields	10^{-2}	10^{-2}	2.10^{-2}
Fresh water at 20℃	3.10^{-3}	3.10^{-3}	5.10^{-3}
Sea water at 20℃	5	5	5

Ground Wave/Radio

- Advantage:
  - have the tendency to bend around the corners or obstructions during propagation
  - not affected by the change in atmospheric conditions
- Disadvantages:
  - High-frequency waves cannot be transmitted as the energy losses are more because of the energy absorption in the earth's atmosphere.
  - These are used to cover short ranges and also involves attenuation of waves as they interact with the eddy currents produced by the surface of the earth.
- Applications:
  - These can be used for **one-way communication** from the military to submerged submarines as they penetrate to a significant depth into seawater
  - AM, FM and television broadcasting can be done with the help of ground waves

ALOHA

- 1970, at the University of Hawaii a method called ALOHA to solve the channel allocation problem is proposed.
- ALOHA system, **uses ground-based radio broadcasting.**
- The idea is applicable to **any system in which uncoordinated users are competing** for the use of a single shared channel.
- Simple, decentralized and works well for low load
  - For many users, max efficiency is 18%
- Two versions of ALOHA are used in Data Networks
  - **Pure and Slotted ALOHA**
  - They differ with respect to whether time is divided into discrete slots into which all frames must fit. Pure ALOHA does not require global time synchronization; slotted ALOHA does.

ALOHA

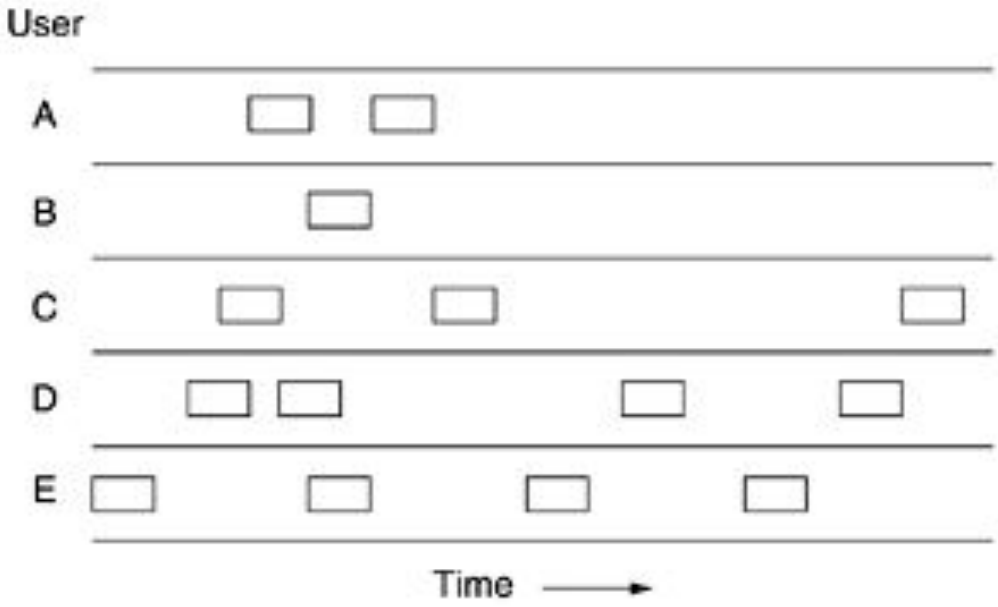
- Systems in which multiple users share a common channel in a way that can lead to conflicts are widely known as **contention** systems.

Pure ALOHA

- The basic idea of an ALOHA system.
  - Let users transmit whenever they have data to be sent.
  - Allow collision, i.e. there will be collisions, of course, and the colliding frames will be damaged.
  - Detect errored frames and **retransmit at random time later**
- However, due to the **feedback property of broadcasting**, a sender can always find out whether its frame was destroyed by listening to the channel, the same way other users do.
    - With a LAN, the feedback is immediate;
    - With a satellite, there is a delay of 270 msec before the sender knows if the transmission was successful.
    - If the frame was destroyed, the sender just **waits a random amount of time** and sends it again. The waiting time must be random or the same frames will collide over and over, in lockstep leading to **Contention**.
  - ALOHA is contention based protocol.

Pure ALOHA

- In pure ALOHA, frames are transmitted at completely arbitrary times.
- A station does not listen to the channel before transmitting, and it has no way of knowing that another frame was already underway.



Pure ALOHA

- Based on statistical probability, the vulnerability period of a frame is measured and overall throughput of the ALOHA system is determined.
- Throughput is calculated by $S=GP_0$, can be
  - Where, S = Throughput, G = Mean (mean no. of frames generated), P_0 = the probability of no other traffic being initiated during the entire vulnerable period
- P_0 can be given by $P_0 = e^{-2G}$. So, throughput can be expressed by,

$$S = Ge^{-2G}$$

- The maximum throughput occurs at $G = 0.5$, with $S = 1/2e$, which is about 0.184. In other words, the best we can hope for a channel utilization is of 18 percent.

Slotted ALOHA

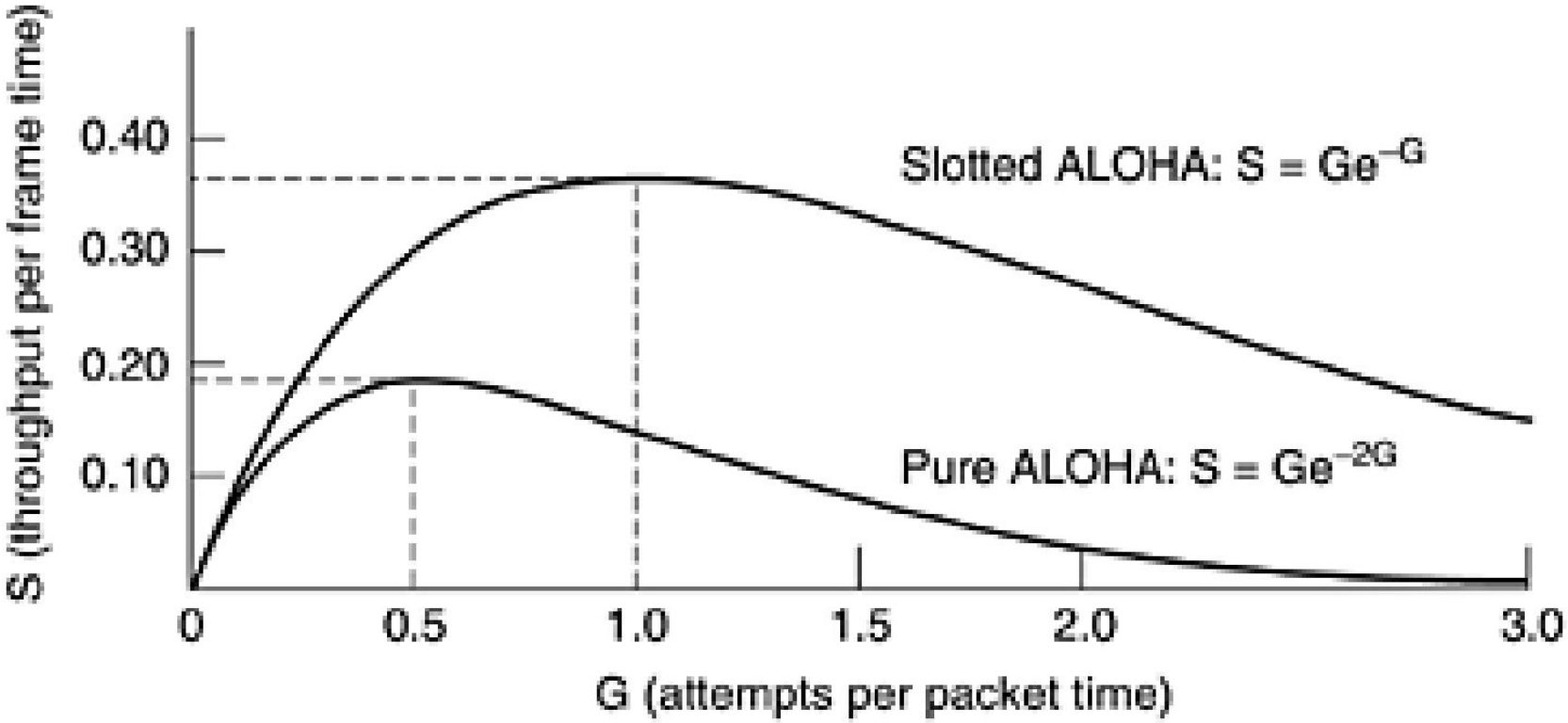
- 1972
- Idea is to divide time into discrete intervals, each interval corresponding to one frame.
  - This approach requires the users to agree on slot boundaries, i.e. to achieve synchronization
    - Can be achieved with one special station emit a pip at the start of each interval, like a clock.
- As in pure ALOHA, a station is not permitted to send whenever it has data. Instead, it is required to wait for the beginning of the next slot.
- Thus, the continuous pure ALOHA is turned into a discrete one.
- Since the vulnerable period is now halved, the probability of no other traffic during the same slot as our test frame is e^{-G} which leads to

$$S = Ge^{-G}$$

Slotted ALOHA

- With slotted ALOHA peaks at $G = 1$, with a throughput of $S = 1/e$ *or* about 0.368, twice that of pure ALOHA.
- If the system is operating at $G = 1$, the probability of an empty slot is 0.368.
- The best we can hope for using slotted ALOHA is 37 percent of the slots empty, 37 percent successes, and 26 percent collisions.
- Operating at higher values of G reduces the number of empties but increases the number of collisions exponentially.

Throughput versus offered traffic for ALOHA systems



CSMA

- Career Sense Multiple Access (CSMA)
- Protocols in which stations listen for a carrier (i.e., a transmission) and act accordingly are called **carrier sense protocols**.
- "Listen-before-talk" policy (CSMA)
  - Good defense against collisions only if "a" is small (LANs)
  - **"a" parameter**: number of packets that fit on the wire
  - a = bandwidth * delay / packet size
  - Small (<<1) for LANs, large (>>1) for satellites
- To deal with Busy (Channel Busy), ***Persistent and Nonpersistent CSMA*** is used.
  - **Persistent** CSMA: retry immediately when channel becomes idle (this may cause instability)
  - **Non persistent CSMA**: retry after random interval of time

CSMA types/variants

Three types of CSMA based on **technique to deal busy**

- **1-persistent CSMA**
  - Wait until idle then go for it
  - Blocked senders can queue up and collide
  - Example: CSMA/CD
- **non-persistent CSMA**
  - Wait a random time and try again
  - Less greedy when loaded, but larger delay
- **p-persistent CSMA**
  - If idle send with probability p until done; assumed slotted time
  - Choose p so that p * # senders < 1; avoids collisions at cost of delay
  - If the station chooses not to transmit (the probability of this event is *1-p*), the sender waits until the next available time slot and transmits again with the same probability *p*.
  - Example: CSMA/CA

CSMA types/variants

Other types of CSMA based on **technique to deal busy**

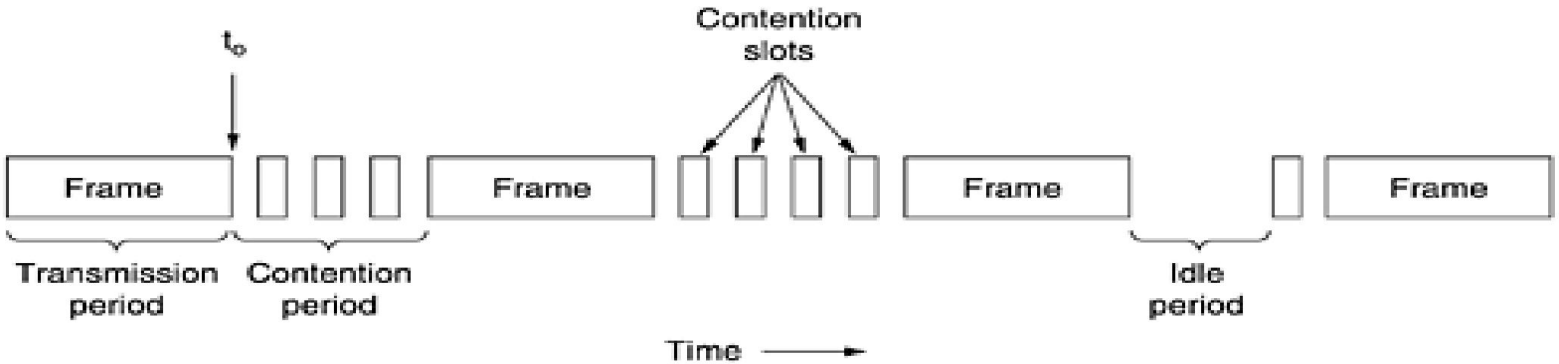
- O-persistent:
  - is used by CobraNet, LonWorks, and the controller area network. This mode assigns a **transmission order to each data node**. When the medium becomes idle, the data node next in line is able to transmit data.

CSMA/CD

- Even with CSMA there can still be collisions. Why?
  - If two stations sense the channel to be idle and begin transmitting simultaneously, they will both detect the collision almost immediately.
  - Rather than finish transmitting their frames (which are irretrievably garbled anyway), they will **abort transmission as soon as the collision is detected.** Quickly terminating damaged frames saves time and bandwidth. This protocol, known as **CSMA/CD** (**CSMA with Collision Detection**) is widely used on LANs in the MAC sublayer.
  - **1-persistent** retransmission is implemented.
- It is the basis of the popular Ethernet LAN

CSMA/CD

- CSMA/CD can be in one of three states: contention, transmission, or idle.
- Because after a station detects a collision,
  - it aborts its transmission, waits a random period of time, and then tries again, assuming that no other station has started transmitting in the meantime.
  - CSMA/CD will consist of alternating contention and transmission periods, with idle periods occurring when all stations are quiet
- Collision detection is an analog process. The station's hardware must listen to the cable while it is transmitting



CSMA/CD

- Collision detection is **easy in wired LANs**:
  - can measure signal strength on the line
- Collision detection **cannot be done in wireless LANs** .
  - CSMA/CA
- CSMA/CD can approach channel utilization =1 in LANs:
  - low ratio of propagation over frame transmission time

Collision-Free Protocols

- Problem in CSMA/CD:
  - Although collisions do not occur with CSMA/CD once a station has unambiguously captured the channel, they can **still occur during the contention period.**
  - These collisions adversely affect the system performance, especially when the cable is long (i.e., large τ) and the frames are short. And because of that CSMA/CD is not universally applicable.
  - CSMA/CD provide no ACK mechanism.
- The protocols that **resolve the contention for the channel without any collisions**, **not even during the contention period** are called Collision-free protocols.

Channel Acquisition Strategies

- Two basic strategies for channel acquisition in a cable network:
  - **contention, as in CSMA, and**
  - **collision-free methods**
- Each strategy can be rated as to how well it does with respect to the two important performance measures, delay at low load and channel efficiency at high load.

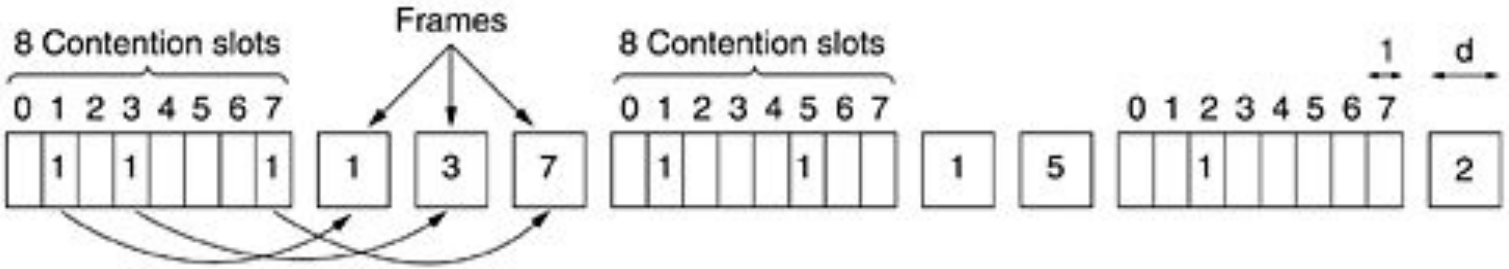
- Under conditions of **light load**, **contention** (i.e., pure or slotted ALOHA) **is preferable** due to its low delay. As the **load increases**, **contention** becomes increasingly **less attractive**, because the overhead associated with channel arbitration becomes greater.
- The reverse is true for the collision-free protocols. At low load, they have high delay, but as the load increases, the channel efficiency improves.

Collision-Free Protocols

- Collision-free protocols
  - Bit-map protocol
  - Binary count down protocol
  - Limited-Contention Protocol
    - Adaptive Tree Walk Protocol
  - WDMA Protocol

Collision-Free Protocols

- **Bit-map protocol**
  - Here the desire/willingness to transmit is broadcast before the actual transmission. That's why these protocols are also called **reservation protocols.**
  - Each contention period consists of exactly *N* slots
  - **Each station has the knowledge thru preparing the Bit-map of willingness.**
  - A station *j* may announce that it has a frame to send by inserting a 1 bit into slot *j.* After all *N* slots have passed by, each station has complete knowledge of which stations wish to transmit. At that point, they begin transmitting in numerical order avoiding collision.
  - If a station becomes ready just after its bit slot has passed by, it is out of luck and must remain silent until every station has had a chance and the bit map has come around again



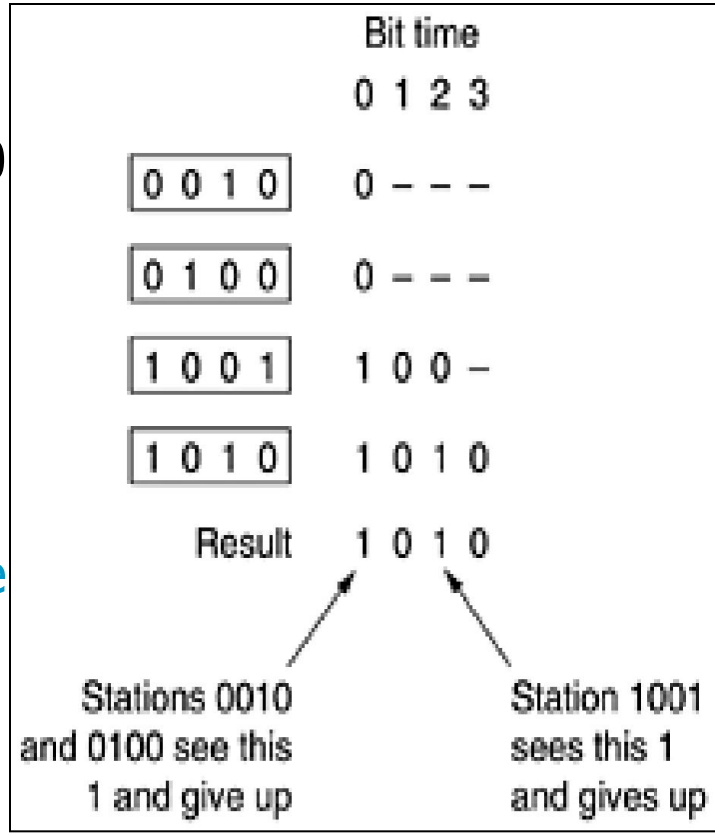
Collision-Free Protocols

- **Disadv:** A problem with the basic bit-map protocol is that the **overhead is 1 bit per station**, so it does not scale well to networks with thousands of stations.

- **Binary count down protocol**
  - Binary count down protocol overcomes the problem of bit-map protocol by using binary station addresses.
  - A station wanting to use the channel **broadcasts its address as a binary bit string**, *starting with the high-order bit*. All addresses are assumed to be the same length. The bits in each address position from different stations are BOOLEAN ORed together.
  - Bidding property:
    - **It has the property that higher-numbered stations have a higher priority than lower-numbered stations, which may be either good or bad, depending on the context.**
  - Conflict avoidance rule:
    - **As soon as a station sees that a high-order bit position that is 0 in its address has been overwritten with a 1, it gives up.**

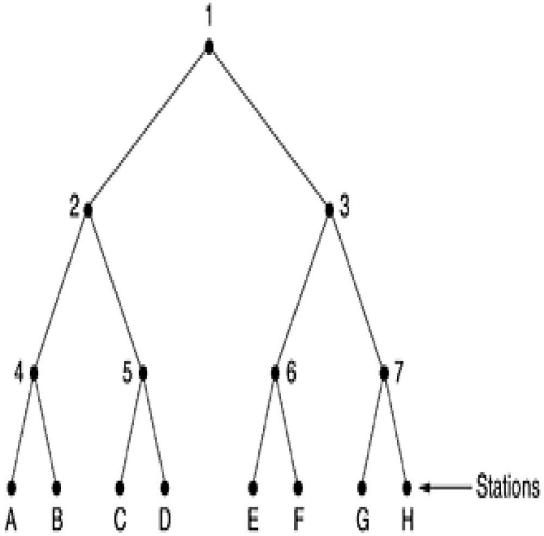
Collision-Free Protocols

- Suppose stations 0010, 0100, 1001, and 1010 are all trying to get the channel,
  - In **the first bit time** the stations transmit 0, 0, 1, and 1, respectively. These are ORed together to form a 1. Stations 0010 and 0100 see the 1 and know that a higher-numbered station is competing for the channel, so they give up for the current round. Stations 1001 and 1010 continue.
  - The next bit is 0, and both stations continue. The next bit is 1, so station 1001 gives up. The winner is station 1010 because it has the highest address. After winning the bidding, it may now transmit a frame, after which another bidding cycle starts.



Collision-Free Protocols

- **Limited-Contention Protocol**
  - It combine the best properties of the contention and collision-free protocols, arriving at a new protocol that uses contention at low load to provide low delay, but uses a collision-free technique at high load to provide good channel efficiency. Such protocols, are called **limited-contention protocols.**
  - Adaptive Tree Walk Protocol
    - The stations are at the leaves of a binary tree. In the first contention slot following a successful frame transmission, slot 0, all stations are permitted to try to acquire the channel. If one of them does so, fine. If there is a collision, then during slot 1 only those stations falling under node 2 in the tree may compete. If one of them acquires the channel, the slot following the frame is reserved for those stations under node 3. If, on the other hand, two or more stations under node 2 want to transmit, there will be a collision during slot 1, in which case it is node 4's turn during slot 2.
- WDMA Protocol
  - Self study



Binary Exponential Backoff Algorithm

- How randomization is done when a collision occurs?
- After a collision, time is divided into discrete slots whose length is equal to the **worst-case round-trip propagation time** on the ether (2τ). To accommodate the longest path allowed by Ethernet, the slot time has been set to 512 bit times, or 51.2 μsec.
- **After the first collision**,
  - Each station waits either 0 or 1 slot times before trying again.
  - If two stations collide and each one picks the same random number, they will collide again.
- **After the second collision,**
  - Each one picks either 0, 1, 2, or 3 at random and waits that number of slot times.
- **If a third collision occurs** (the probability of this happening is 0.25), then the next time the number of slots to wait is chosen at random from the interval 0 to $2^3 - 1$.
- In general, after *i* collisions, a random number between 0 and $2^i - 1$ is chosen, and that number of slots are skipped. However, after ten collisions have been reached, the randomization interval is frozen at a maximum of 1023 slots. After 16 collisions, the controller throws in the towel and reports failure back to the computer. Further recovery is up to higher layers.

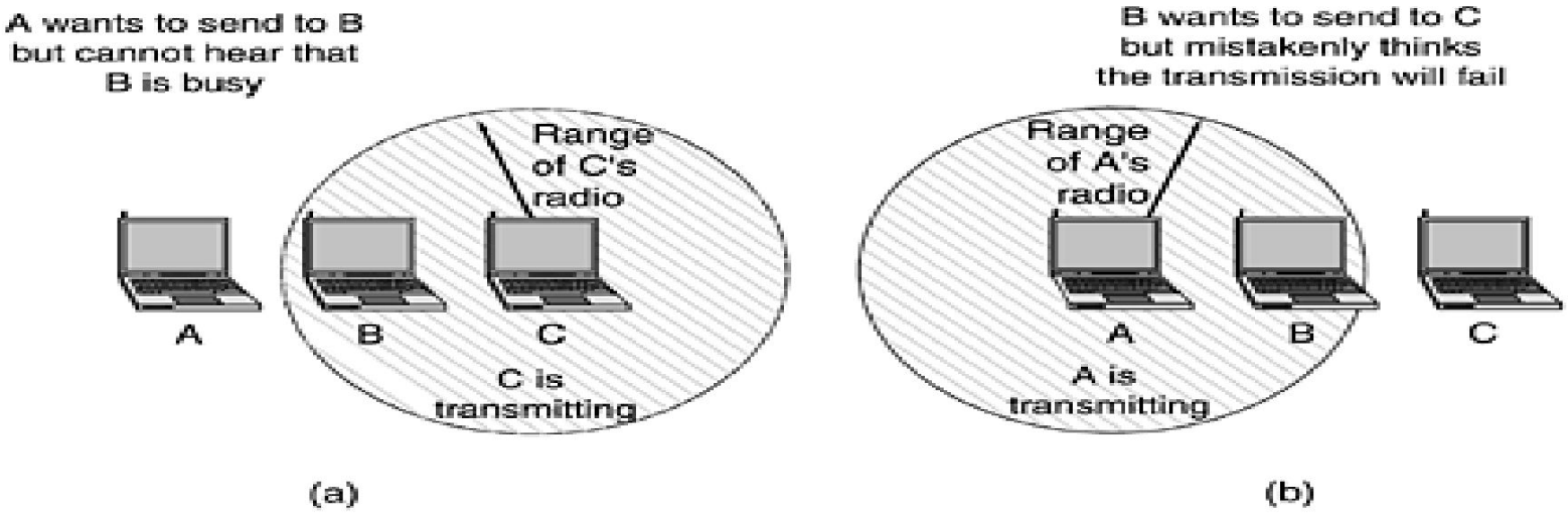
Binary Exponential Backoff Algorithm

- This algorithm, called **binary exponential backoff**, was chosen to dynamically adapt to the number of stations trying to send.
- If the randomization interval for all collisions was 1023, the chance of two stations colliding for a second time would be negligible,
- But the average wait after a collision would be hundreds of slot times, introducing significant delay.
- On the other hand, if each station always delayed for either zero or one slots, then if 100 stations ever tried to send at once, they would collide over and over until 99 of them picked 1 and the remaining station picked 0. This might take years. **By having the randomization interval grow exponentially as more and more consecutive collisions occur, the algorithm ensures a low delay when only a few stations collide but also ensures that the collision is resolved in a reasonable interval** when many stations collide. Truncating the backoff at 1023 keeps the bound from growing too large.

CSMA/CA

- CSMA with Collision Avoidance
- Since we can't detect collisions, we avoid them
  - CSMA/CA as opposed to CSMA/CD
  - Not greedy like Ethernet
- Collision Avoidance (CA) is used to improve CSMA performance by not allowing wireless transmission of a node if another node is transmitting, thus reducing the probability of collision due to the use of a random truncated binary exponential backoff time.
- **CS:** listen before transmitting.
  - When medium busy, choose random backoff interval
  - Wait for that many idle timeslots to pass before sending
- **CA**: transmit short "jamming" signal before sending frame
  - It essentially reserves medium, let's other stations know about intent to transmit
- Collisions can be inferred
  - Use CRC and ACK from receiver to infer "no collision"
  - on collision, binary exponential backoff like Ethernet

Wireless MAC problems



- **Hidden Terminal/Station Problem**
  - A station can't detect a potential competitor for the medium because the competitor is too far away/out of range. Fig(a)
- **Exposed Terminal/Station Problem**
  - As shown in Fig(b).

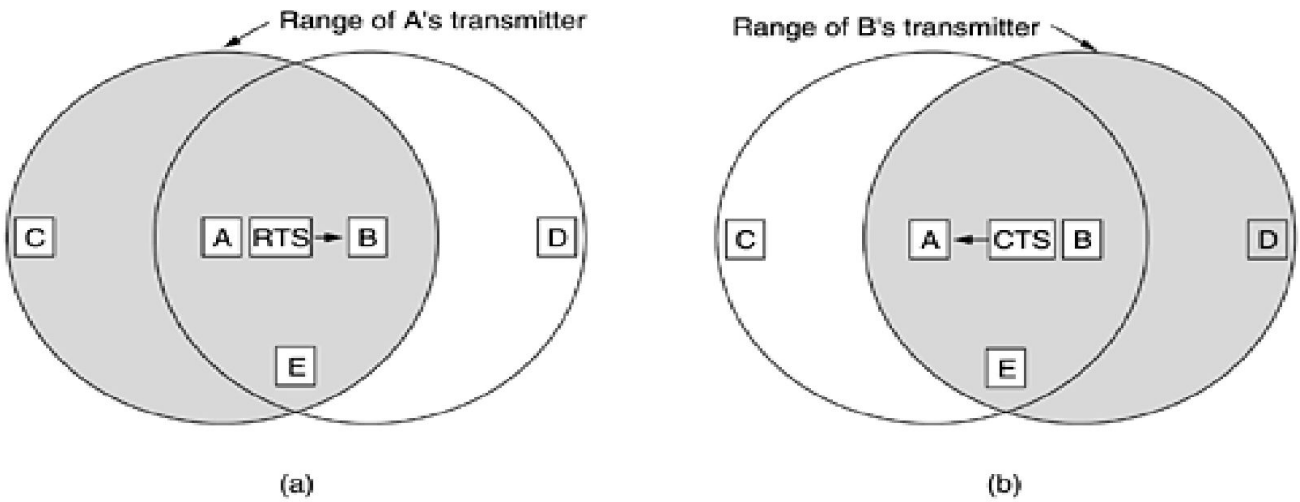
Wireless LAN protocols

Based on CSMA/CA with modification:

- MACA
  - For wireless LANs **MACA** (**Multiple Access with Collision Avoidance**) (Karn, 1990) is proposed. The basic idea behind it is for the sender to stimulate the receiver into outputting a short frame (RTS), so that stations nearby can detect this transmission and avoid transmitting for the duration of the upcoming (large) data frame.
  - RTS (request to send) and CTS (Clear to send) short frames are used. 30 bytes long frames
  - Disadv: Lost frames are not retransmitted
- MACAW
  - Bharghavan et al. (1994) fine tuned MACA to improve its performance and renamed their new protocol **MACAW** (**MACA for Wireless**).
  - It is noticed that without data link layer acknowledgements (ACK), lost frames were not retransmitted until the transport layer noticed their absence, much later.

MACA and MACAW

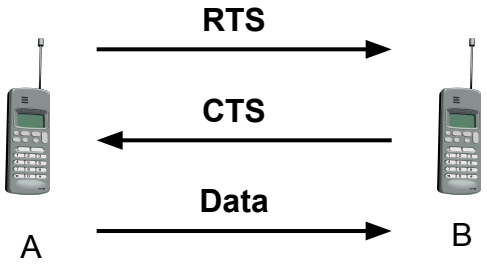
- **MACA Protocol**
  - *C* is within range of *A* but not within range of *B*. Therefore, it hears the RTS from *A* but not the CTS from *B*. As long as it does not interfere with the CTS, it is free to transmit while the data frame is being sent. In contrast, *D* is within range of *B* but not *A*. It does not hear the RTS but does hear the CTS. Hearing the CTS tips it off that it is close to a station that is about to receive a frame, so it defers sending anything until that frame is expected to be finished. Station *E* hears both control messages and, like *D*, must be silent until the data frame is complete.
  - Despite these precautions, collisions can still occur. For example, *B* and *C* could both send RTS frames to *A* at the same time. These will collide and be lost. waits a random amount of time and tries again later (binary exponential backoff)



MACA and MACAW

- MACAW
  - **without data link layer acknowledgements, lost frames were not retransmitted until the transport layer noticed their absence, much later**. They solved this problem by introducing an ACK frame after each successful data frame.
  - They also observed that CSMA has some use, namely, to keep a station from transmitting an RTS at the same time another nearby station is also doing so to the same destination, so carrier sensing was added. In addition, they decided to run the backoff algorithm separately for each data stream (source-destination pair), rather than for each station. This change improves the fairness of the protocol. Finally, they added a mechanism for stations to exchange information about congestion and a way to make the backoff algorithm react less violently to temporary problems, to improve system performance.

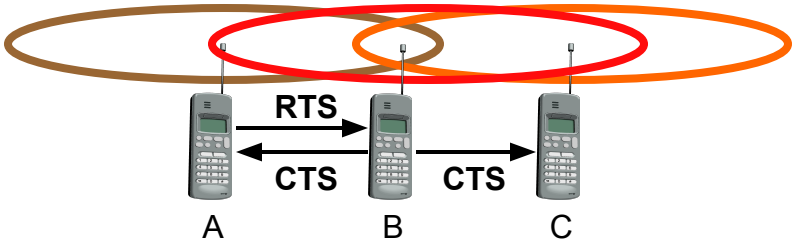
An Illustration: 3-way handshake



Can it solve the hidden terminal problem?

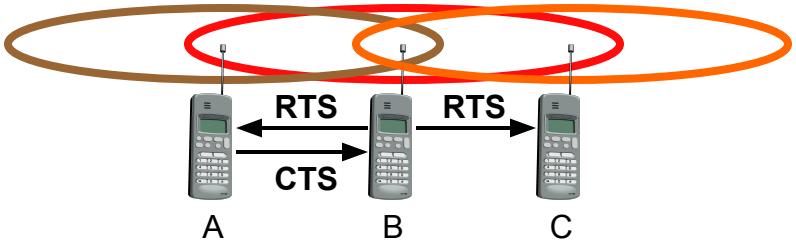
Can it solve the exposed terminal problem?

A Solution: Hidden Terminal



- MACA avoids the hidden terminal problem
  - Both A and C want to send to B
  - A sends RTS first
  - C waits after receiving CTS from B

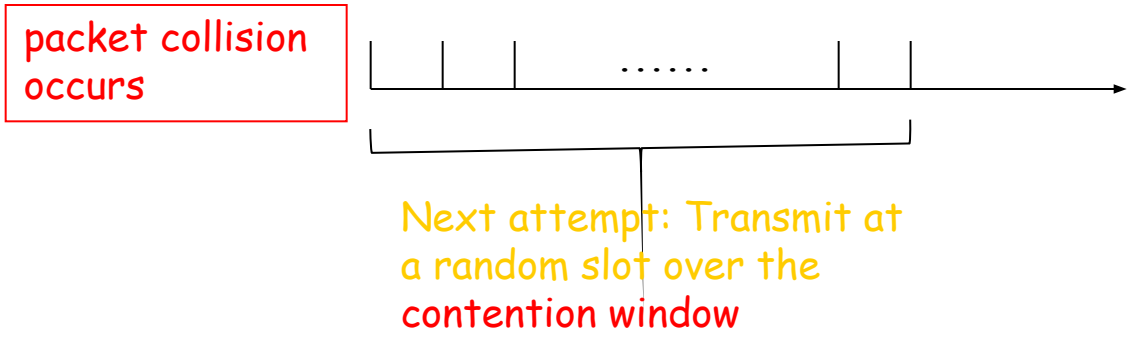
A Solution: Exposed Terminal



- MACA avoids the exposed terminal problem
  - B wants to send to A, while C to another terminal
  - now C does not have to wait, for it cannot
    receive CTS from A

Packet Collision

- Collisions may occur during RTS-CTS exchange.



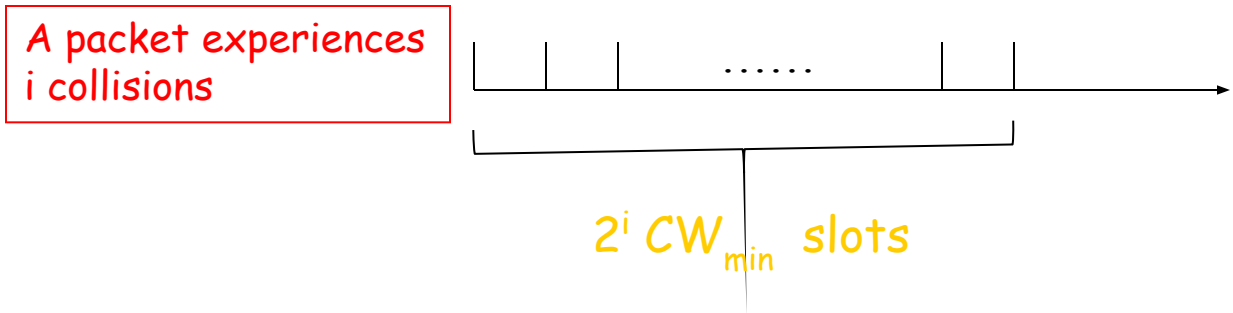
How large is the contention window?

Binary Exponential Backoff

- The contention window size is adjusted dynamically.
  - binary exponential backoff is used.
- When a terminal fails to receive CTS in response to its RTS, it increases the contention window
  - *cw* is doubled (up to an upper bound, CW_{max})
- When a node successfully completes a data transfer, it restores *cw* to CW_{min}

Binary Exponential Backoff

- The contention window size is doubled whenever a collision occurs.



IEEE 802.11 MAC Protocols

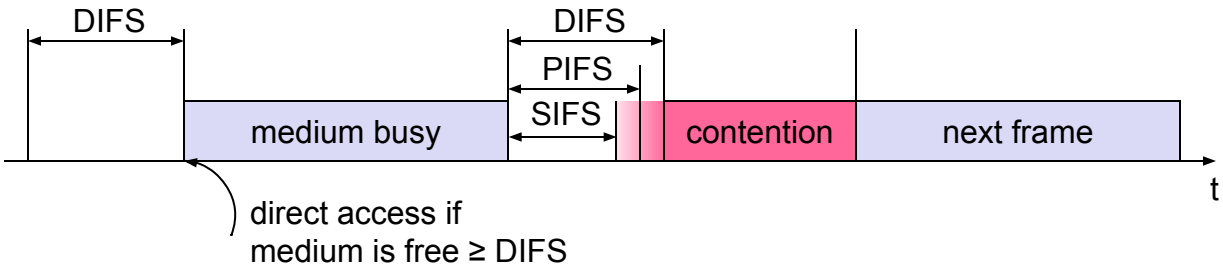
- Two traffic services are supported
  - Asynchronous Data Service
    - Best-effort services
  - Time-bounded Service (optional)
    - Guarantee a maximum delay
    - Available only in infrastructure mode

Two Classes of Access Mechanisms

- Distributed Coordination Function (DCF)
  - Support asynchronous data services
  - CSMA/CA (with minor modification)
  - CSMA/CA with RTS/CTS exchange (optional)
- Point Coordination Function (PCF) (optional)
  - Support time-bounded services
  - Polling from AP

Inter-Frame Spacings (for waiting time before medium access)

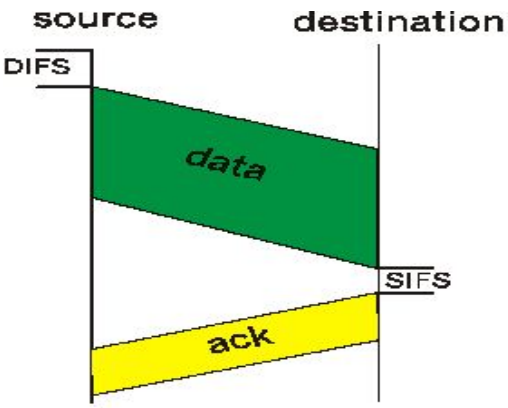
- SIFS (Short Inter Frame Spacing)
  - highest priority, for ACK, CTS, polling response
- PIFS (PCF IFS)
  - medium priority, for time-bounded service using PCF
- DIFS (DCF IFS)
  - lowest priority, for asynchronous data service



Method 1a: CSMA/CA

802.11 CSMA/CA: sender

- if sense channel idle for **DIFS** sec.(plus a RBT if the medium was busy)
  then transmit entire frame (no collision detection)
- if sense channel busy
  then wait for the channel to become idle for DIFS and choose a random backoff time and countdown the backoff timer for every free slot



DIFS: Distributed Inter Frame Spacing

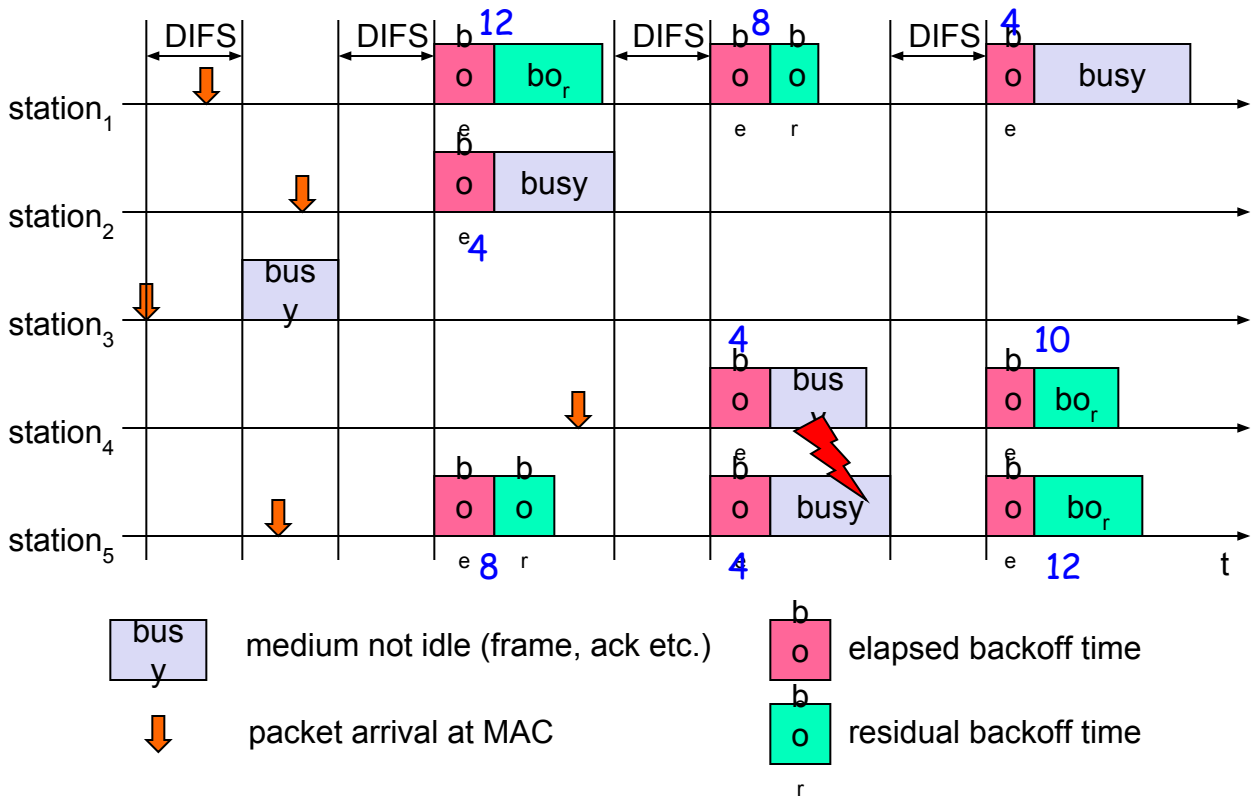
SIFS: Short Inter Frame Spacing

802.11 CSMA/CA: receiver

if received OK

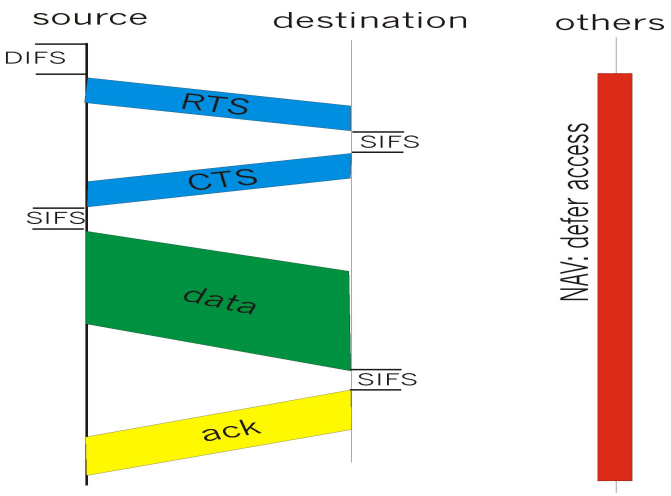
return ACK (16 bytes) after **SIFS**

Example (with backoff timer)



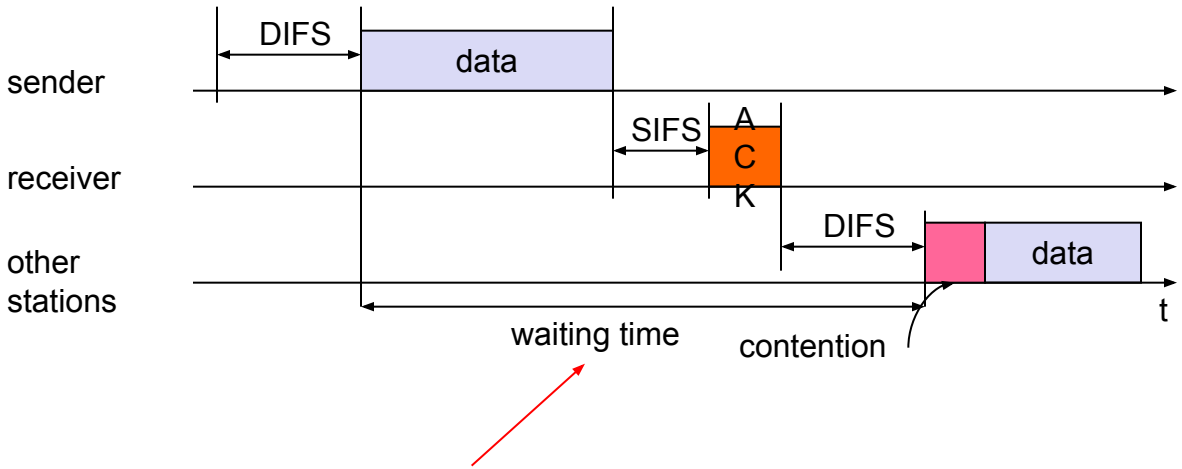
Method 1b: CSMA/CA with RTS-CTS

- CSMA/CA: explicit channel reservation
  - sender: send RTS (20 bytes)
  - receiver: reply with CTS (16 bytes)
- CTS reserves channel for sender, notifying (possibly hidden) terminals



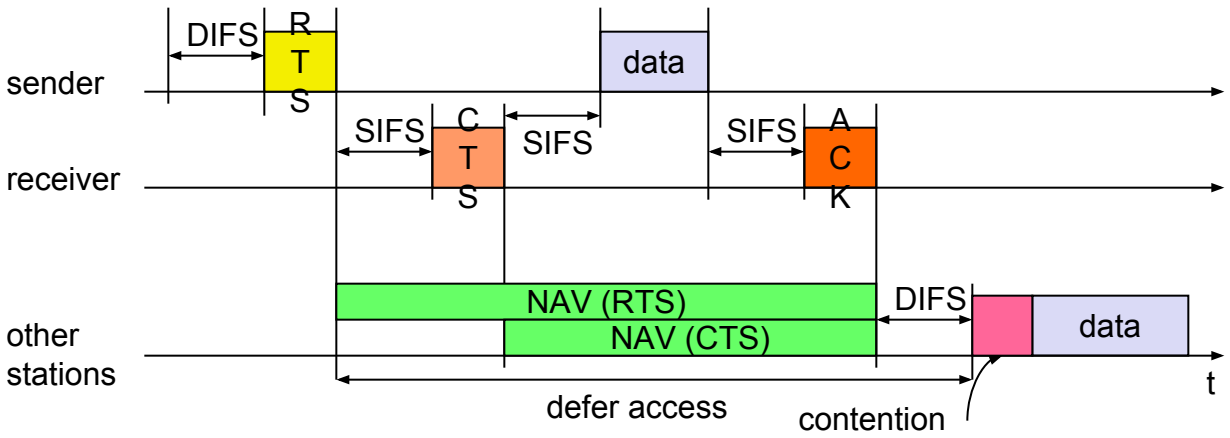
4-way handshake

No Collision during Data Transmission



How can other stations know how long the waiting time is?

Net Allocation Vector (reserves the medium for one sender exclusively)



The RTS packet has a duration field, which consists of information about the length of data packet.

Other stations hear the RTS packet set their NAV accordingly.

The CTS packet also has the duration field.

Other stations hear the CTS packet set their NAV accordingly.

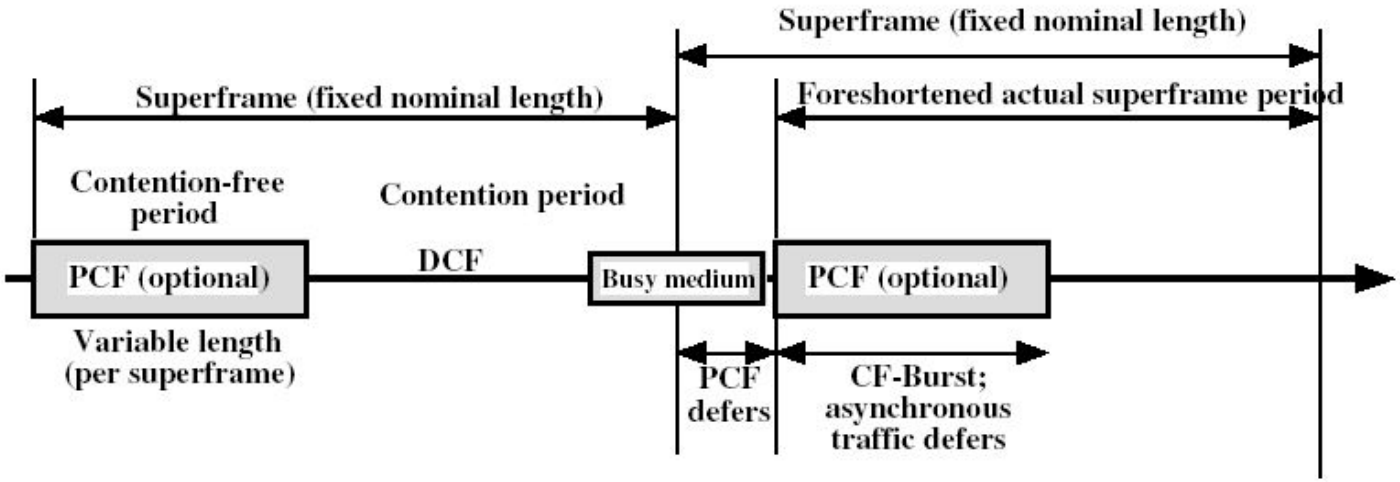
Using RTS/CTS

- Removes hidden terminal problem
- Collision may occur at the beginning of transmission (RTS/data packets)
- Non-negligible overhead (b/w and delay)
- RTS Threshold to determine when to use/disable RTS/CTS option (based on frame size)

Method 2: Point Coordination Function

- Polling by the access point (or point coordinator)
- Sends polling message after waiting for PIFS
- Since PIFS is smaller than DIFS, it can lock out all asynchronous traffic
  - To prevent this, an interval called superframe is defined.

Two parts of a Superframe



Contention-free Period:

The point coordinator polls stations with time-bounded service in a round-robin fashion

Contention Period:

The point coordinator idles for the remainder of the superframe, allowing for asynchronous access.

Ethernet

- 1980s
- IEEE 802.3 standard wired LAN (1-persistent CSMA/CD) is known as Ethernet.
- This is generally a LAN technology with some WAN applications. Physical connections are made between nodes and/or infrastructure devices (hubsThis is generally a LAN technology with some WAN applications. Physical connections are made between nodes and/or infrastructure devices (hubs, switchesThis is generally a LAN technology with some WAN applications. Physical connections are made between nodes and/or infrastructure devices (hubs, switches, routers) by various types of copper or fiber cable.
- Uses 48 bit long MAC address
- Ethernet divide a stream of data into individual packets called frames of size 1500bytes. Each frame contains source and destination addresses and error-checking data so that damaged data can be detected and re-transmitted.
- Many types :
  - Classic Ethernet (10BASEx): 10 Mbps over coaxial cable
    - baseband signals, Manchester encoding, preamble, 32 bit CRC
  - Newer versions are much faster;, 100BASEx, 1000BASEx
    - Fast (100 Mbps), Gigabit (1 Gbps)

Common kinds of Ethernet cabling

Name	Cable	Max. seg.	Nodes/seg.	Advantages
10Base5	Thick coax	500 m	100	Original cable; now obsolete
10Base2	Thin coax	185 m	30	No hub needed
10Base-T	Twisted pair	100 m	1024	Cheapest system
10Base-F	Fiber optics	2000 m	1024	Best between buildings

Ethernet

- The 48 bit Frame format :

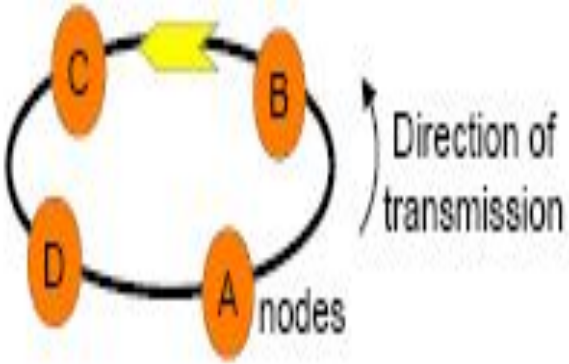
Preamble (8)	Dest (6)	Source (6)	Type (2)	Payload (var)	Pad (var)	CRC (4)

- Min frame 64 bytes, max 1500 bytes
- Max length 2.5 km, max between stations 500m (repeaters)
- Addresses unique per adaptor; 6 bytes; globally assigned
- Broadcast media is readily tapped:
  - Promiscuous mode (i.e. pass all traffic it receives); multicast addresses
- **Binary Exponential Backoff is used by Ethernet:**
  - Build on 1-persistent CSMA/CD
  - On collision: jam and exponential backoff
    - **Jamming: send 48 bit sequence to ensure collision detection**
  - Backoff:
    - **First collision: wait 0 or 1 frame times at random and retry**
    - **Second time: wait 0, 1, 2, or 3 frame times**
    - **N th time (N<=10): wait 0, 1, …, 2^N-1 times**
    - **Max wait 1023 frames, give up after 16 attempts**
    - **Scheme balances average wait with load**

Token Bus

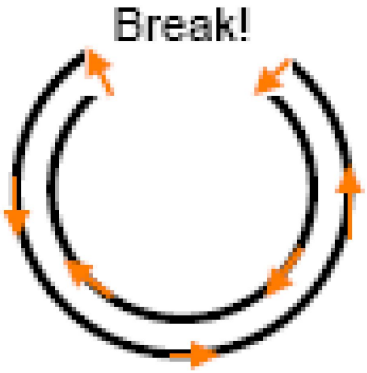
- IEEE standard 802.4
- Token bus is a network implementing the token ring protocol over a **"virtual ring"** on a coaxial cable.
- A **token** is passed around the network nodes and only the node possessing the token may transmit.
  - If a node doesn't have anything to send, the token is passed on to the next node on the virtual ring.
  - Each node must know the address of its neighbour in the ring, so a special protocol is needed to notify the other nodes of connections to, and disconnections from, the ring.
- **Disadvantage:**
  - It is difficult to handle device failures and adding new stations to a network,
  - Unreliable and difficult to upgrade

Token Ring



- IEEE 802.5 standard
- Nodes form a topology of ring
- A special **three-byte frame called a token** that travels around the ring.
- Token-possession grants the possessor node permission to transmit on the medium.
- Token ring frames travel completely around the loop.
- Token rotates permission to send around node
- Sender injects packet into ring and removes later
  - Maximum token holding time (THT) bounds access time
  - Early or delayed token release
  - Round robin service, acknowledgments and priorities
- Monitor nodes ensure health of ring

FDDI (Fiber Distributed Data Interface)



- Run on fiber
- Implements **dual ring**
  - two independent rings transmitting data in opposite direction
  - second is not used for normal operation but used only if primary fails
  - stations may be allowed to connect on a single cable witn single attachment station (SAS). Uses concentrator to connect several SASs to dual ring.
- Roughly a large, fast token ring
  - 100 Mbps and 200km vs 4/16 Mbps and local
  - Dual counter-rotating rings for redundancy
  - Complex token holding policies for voice etc. traffic
- Token ring advantages
  - No contention, bounded access delay
  - Supports fair, reserved, priority access
- Disadvantages
  - Complexity, reliability, scalability

WLAN (Wireless LAN)

- IEEE 802.11x standard is known as WLAN.
- Uses CSMA/CA with RTS (Request to Send) CTS (Clear to Send) method. RTS/CTS packet size threshold is 0–2347 octets.
- A node (sender) wish to send data asks permission to use the medium by sending a **Request to Send** (RTS) packet.
- A **Clear to Send** (CTS) packet sent by the intended receiver R allows the sender node.
- Eventually this (RTS/CTS exchange) alerts all nodes within range of the sender, receiver or both, to not transmit for the duration of the main transmission.
- Any other node receiving the RTS or CTS frame should refrain from sending data for a given time . The amount of time the node should wait before trying to get access to the medium is included in both the RTS and the CTS frame. This protocol was designed under the assumption that all nodes have the same transmission ranges.
- Therefore, RTS/CTS is an additional method to implement ***virtual carrier sensing*** **in CSMA/CA**. By default, 802.11 (i.e. WLAN) relies on *physical carrier sensing* only which is known to suffer from the hidden node problem.

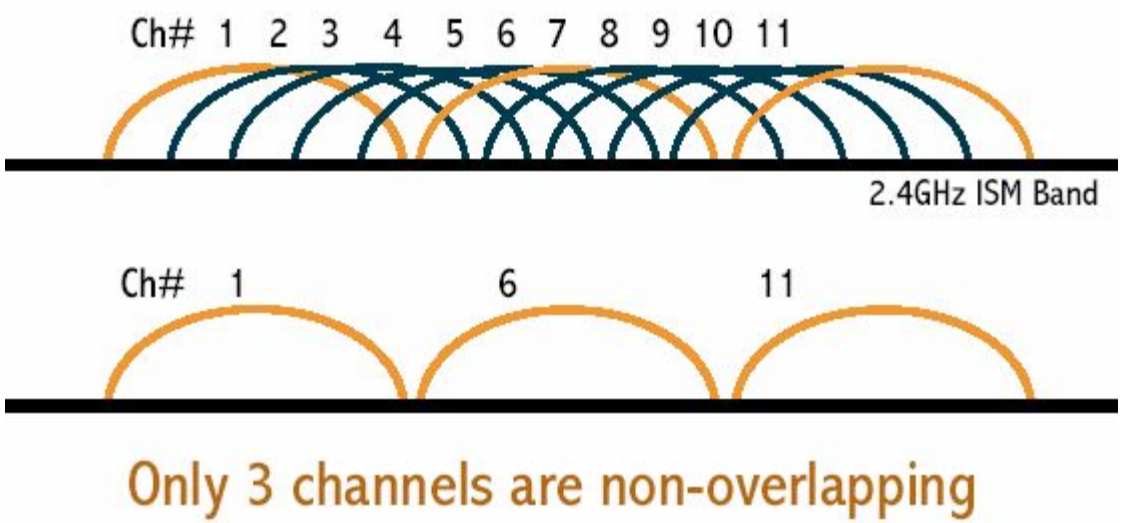
WLANs

Types of WLANs :

- **802.11a**
  - 5GHz range
  - Up to 54Mbps, uses OFDM
- **802.11b**
  - 2.4GHz unlicensed radio spectrum
  - Up to 11Mbps, uses DSSS
- **802.11g**
  - 2.4 GHz
  - Up to 54Mbps, uses OFDM
- **802.11n**
  - 2.4 GHz and 5 GHz
  - Up to 600Mbps, uses OFDM and MIMO
- All use CSMA/CA for multiple access
- Options RTS/CTS
- All have base-station and ad-hoc network versions

802.11 Channels

Channel Scheme used in US and India



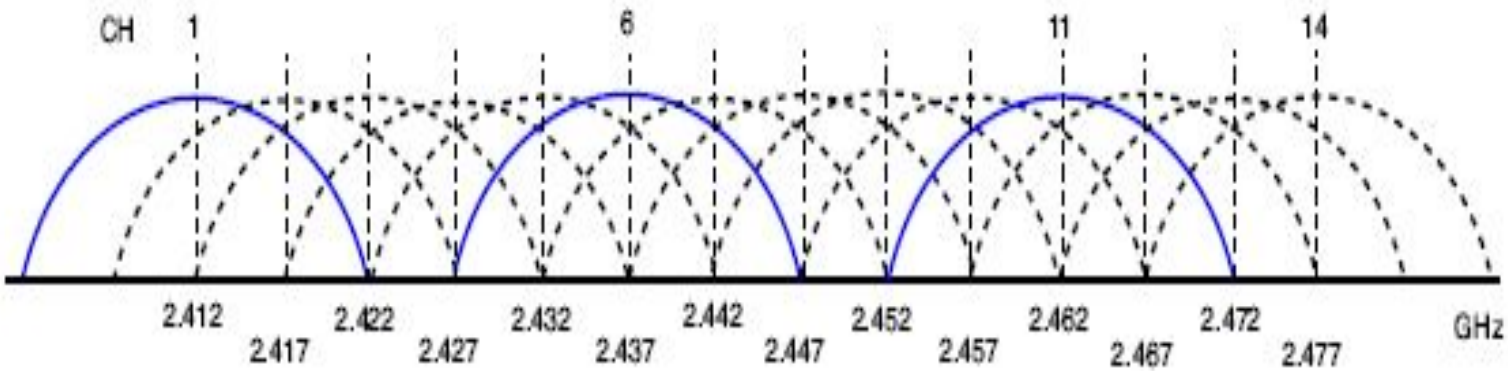


Fig. 2. 802.11 channels in the 2.4 GHz ISM band.

LLC

- IEEE 802.2

- It defines LLC, which is the upper portion of the data link layer of the OSI model.
- LLC sublayer presents a **uniform interface to the user** of the data link service, usually the n/w layer.
- Provides **two connectionless** and **one connection-oriented** operational modes
- **Type 1** is an unacknowledged connectionless mode for a datagram service. It allows for sending frames
  - to a single destination (point-to-pointto a single destination (point-to-point or unicast transfer),
  - to multiple destinations on the same network (multicast),
  - or to all stations of the network (broadcast).

LLC

- The use of multicasts and broadcasts reduce network traffic when the same information needs to be propagated to all stations of the network.
- The Type 1 service provides **no guarantees regarding the order of the received frames** compared to the order in which they have been sent; the sender does not even get an acknowledgment that the frames have been received.
- **Type 2** is a connection-oriented operational mode.
  - Sequence numbering ensures that the frames received are guaranteed to be in the order they have been sent, and no frames are lost.
- **Type 3** is an acknowledged connectionless service. It supports point-to-point communication only.

LLC header

- The 802.2 header includes
  - two eight-bit address fields, called **service access points** or SAPs. They are DSAP, and SSAP.
    - The low-order bit of the DSAP indicates whether it contains an individual or a group address. If the low-order bit is 0, the remaining 7 bits of the DSAP specify an individual address, which refers to a single local service access point (LSAP) to which the packet should be delivered.
    - If the low-order bit is 1, the remaining 7 bits of the DSAP specify a group address, which refers to a group of LSAPs to which the packet should be delivered.
    - The low-order bit of the SSAP indicates whether the packet is a command or response packet; if it's 0, the packet is a command packet, and if it's 1, the packet is a response packet. The remaining 7 bits of the SSAP specify the LSAP from which the packet was transmitted.

VLAN

- **IEEE 802.1Q standard**
- Description:
  - Group of devices on one or more **physical LANs** that are configured as if they are **logically attached to the same wire**
  - LAN's based on Logical instead of Physical connections
- Used to help alleviate traffic congestion without adding more bandwidth
- **Used to separate out users into logical groups of workers, regardless of actual physical location.**
- Usage scenarios:
  - Say you want workers assigned to the same project to be grouped logically together for control of traffic but they are physically located in different physical areas
  - Say you want to divide up the broadcast domain in a large flat network without using a bunch of routers
- **Must be supported by the switch:** switches must have the ability to support more than one subnet

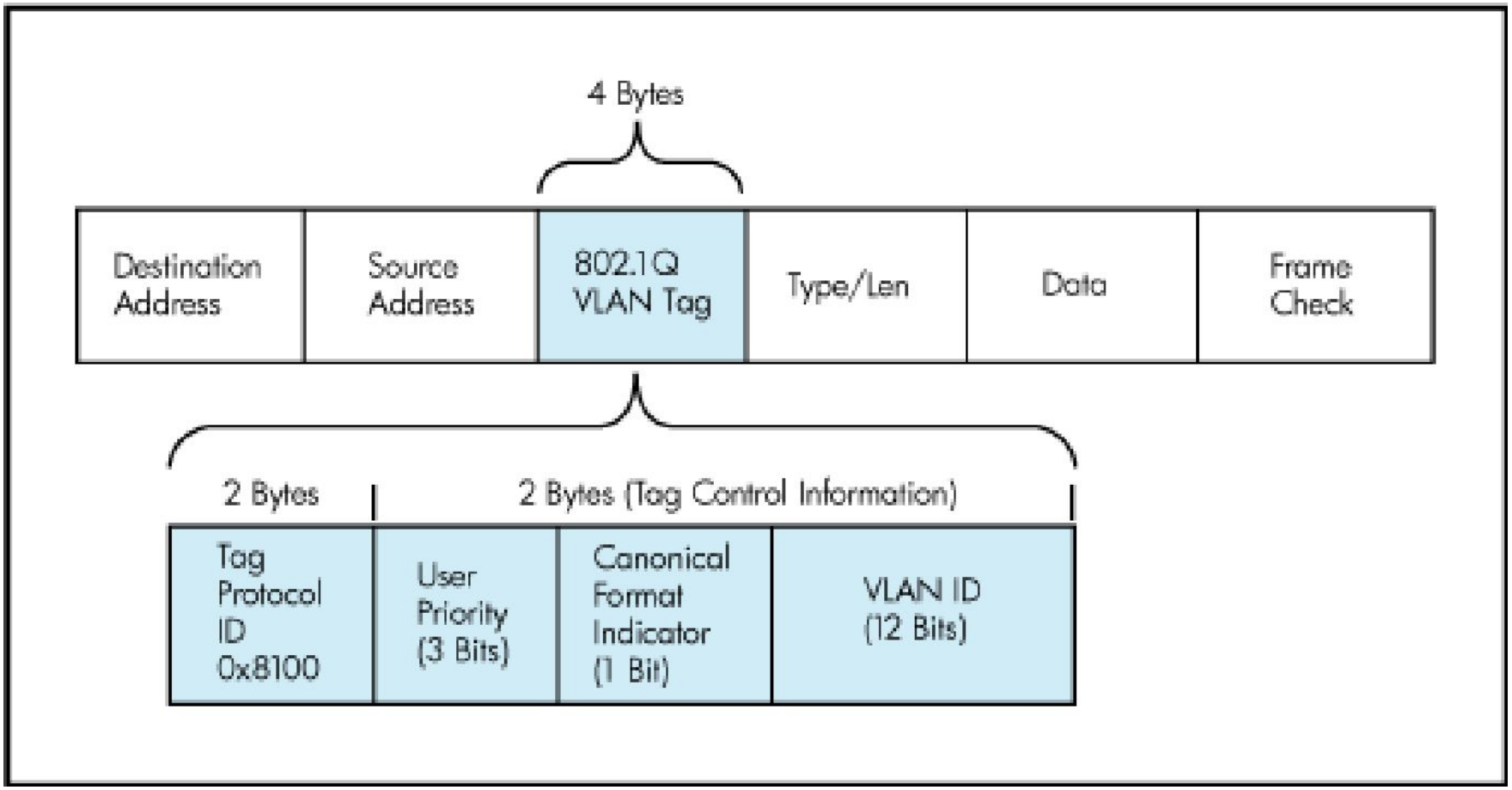
VLAN Types and Strategies

- Types:
  - **Port Based** - Configured at each switch port
    - Port X on Switch A belongs to VLAN 1
    - Port Y on Switch B belongs to VLAN 1
  - **MAC Address Based** - Uses the end stations MAC address for VLAN Assignment
    - Host X MAC belongs to VLAN 1
    - Host Y MAC belongs to VLAN 1
  - **Protocol Based** - Uses LAN protocol to determine VLAN assignment
    - Host X uses IP and belongs to VLAN 1
    - Host Y uses IPX and belongs to VLAN 2
  - **Dynamic Based** - A User Profile (stored as a database) determine VLAN assignment
    - When Host X logs in, the profile says to connect the user to VLAN 1
- Strategies:
  - At the User Level
  - At the Wiring Closet Level
  - At the Distribution Switch Level

VLAN

- How does the network know where to send the data when a VLAN has been put into place?
  - Initially, IEEE 802.10, "Interoperable LAN/MAN Security (SILS)" standard released in 1992, **puts the** VLAN information between the MAC and LAN address space. Today, **using IEEE 802.1Q, Frames can be tagged with the VLAN information**.
- How do we interconnect these smart switches?
  - We could try to manually connect them together.
  - This is not recommended. One can easily make a mistake when physically interconnecting the appropriate ports for the VLAN.

802.1Q Tag


4 Bytes
Destination Address
Source Address
802.1Q VLAN Tag
Type/Len
Data
Frame Check
2 Bytes
2 Bytes (Tag Control Information)
Tag Protocol ID 0x8100
User Priority (3 Bits)
Canonical Format Indicator (1 Bit)
VLAN ID (12 Bits)

802.1Q Terminology

- **access port / link** - defines a port with one or more untagged VLANs, and a link connecting two such ports.
- **trunk port / link** - defines a port with multiple VLANs that are all tagged, and a link connecting two such ports.
- **hybrid port / link** - defines a port with both untagged and tagged VLANs, and a link connecting two such ports.
- **VID** - VLAN ID
- **PVID** - Port VLAN ID
- **tagged frame** - An Ethernet (IEEE 802.3) frame with the 802.1Q tag.
- **clear frame** - An Ethernet frame with no tag.
- **VLAN trunking** - a generic networking term to describe the process of forwarding multiple VLANs across a single link, whether via 802.1Q or proprietary protocols like Cisco's InterSwitch Link Protocol (ISL).

WPAN or PAN

- IEEE 802.15 is Bluetooth
- IEEE 802.15.4 standard is known as Low Rate Wireless Personal Area N/W.

- Self study

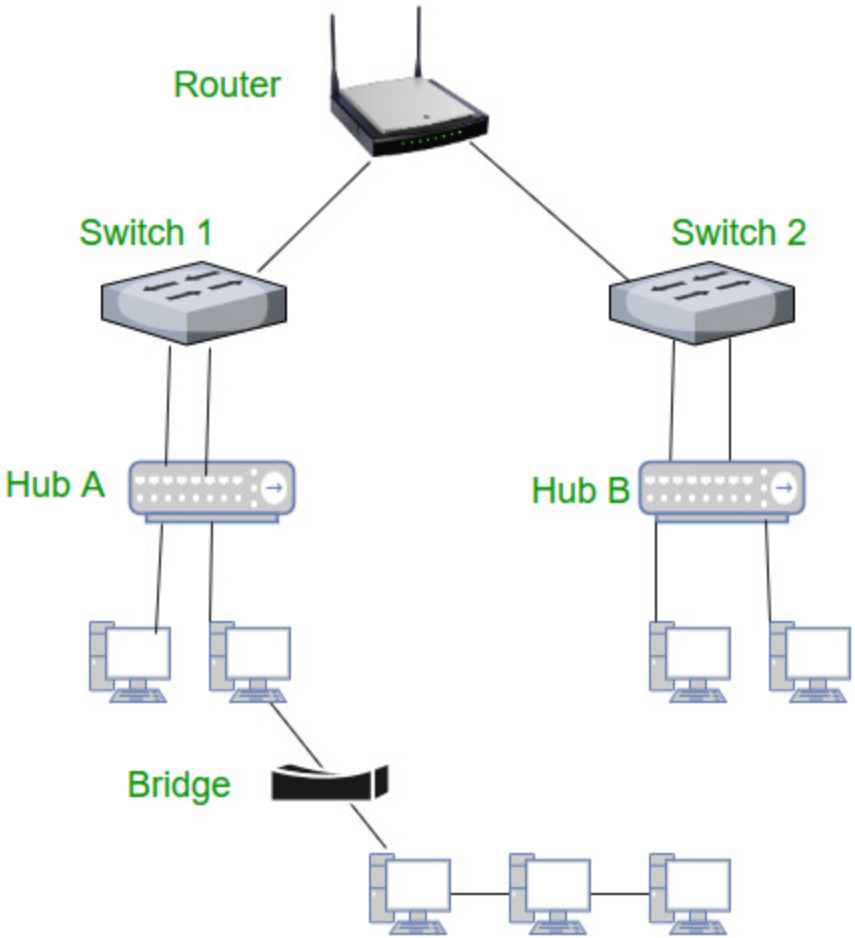
WiMax

- Worldwide Interoperability for Microwave Access
- 802.16
- original IEEE 802.16 standard (now called "**Fixed WiMAX**") was published in 2001
- **Mobile WiMAX** (originally based on **802.16e,** 2005) is the revision that was deployed in many countries, and basis of future revisions such as 802.16m-2011.
- WiMAX is sometimes referred to as "**Wi-Fi on steroids”**. and can be used for a number of applications including broadband connections, cellular backhaul, hotspots, etc. It is similar to Wi-Fi, but it can enable usage at much greater distances.[5]
- Self study

Network Devices

- In an Ethernet network there are 4 devices that from the outside look very similar
  - Hub,
  - Repeater,
  - Bridge,
  - Switch,
  - Router,
  - Gateways and
  - Brouter

Network Devices



Network Devices

- **Repeater** –
  - Operates at the physical layer (i.e. L1)
  - Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network
  - It do not amplify the signal. When the signal becomes weak, they copy the signal bit-by-bit and regenerate it at the original strength
  - It is a 2 port device

Network Devices

- **Hub** –
  - A hub is basically a multiport repeater.
  - A hub connects multiple wires coming from different branches, For example: the connector in star topology which connects different stations.
  - Hubs cannot filter data, so data packets are sent to all connected devices. In other words, collision domain of all hosts connected through Hub remains one.
  - Hubs do not have intelligence to find out best path for data packets which leads to inefficiencies and wastage

Network Devices

- **Types of Hub**
  - **Active Hub**
    - These are the hubs which have their own power supply and can clean, boost and relay the signal along the network.
    - It serves both as a repeater as well as wiring center. These are used to extend maximum distance between nodes.
  - **Passive Hub**
    - These are the hubs which collect wiring from nodes and power supply from active hub.
    - These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend distance between nodes.

Network Devices

- **Bridge** –
  - A bridge operates at data link layer (i.e. L2)
  - A bridge is a repeater, with add on functionality of filtering content by reading the MAC addresses of source and destination.
  - It is also used for interconnecting two LANs working on the same protocol.
  - It has a single input and single output port, thus making it a 2 port device.

Network Devices

- **Types of Bridges**
  - **Transparent Bridges**
    - These are the bridges in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary.
    - These bridges makes use of two processes i.e. bridge forwarding and bridge learning.
  - **Source Routing Bridges**
    - In these bridges, routing operation is performed by source station and the frame specifies which route to follow.
    - The host can discover path by sending a special frame called discovery frame, which spreads through the entire network using all possible paths to destination.

Network Devices

- **Switch** –
  - Switch is data link layer device (i.e. L2)
  - A switch is a multi port bridge with a buffer and a design that can boost its efficiency (large number of ports imply less traffic) and performance.
  - Switch can perform error checking before forwarding data, that makes it very efficient as it does not forward packets that have errors and forward good packets selectively to correct port only. In other words, switch divides collision domain of hosts, but broadcast domain remains same.

Network Devices

- **Routers** –
  - Router is mainly a Network Layer device (i.e. L3)
  - A router is a device like a switch that routes data packets based on their IP addresses.
  - Routers normally connect LANs and WANs together
  - Have a dynamically updating routing table based on which they make decisions on routing the data packets.
  - Router divide broadcast domains of hosts connected through it.

Network Devices

- **Gateway** –
  - Operate at any network layer (i.e. L3)
  - A gateway, as the name suggests, is a passage to connect two networks together that may work upon different networking models.
  - They basically works as the messenger agents that take data from one system, interpret it, and transfer it to another system.
  - Gateways are also called protocol converters. Gateways are generally more complex than switch or router.

Network Devices

- **Brouter** –
  - It can work either at data link layer (i.e. L2) or at network layer (i.e. L3).
  - It is also known as bridging router is a device which combines features of both bridge and router.
  - Working as router, it is capable of routing packets across networks and working as bridge, it is capable of filtering local area network traffic.

Required Reading

- Tanenbaum chapter